

**EEHC DISTRIBUTION MATERIALS SPECIFICATION
HYDRAULIC CRIMPING TOOL FOR LOW AND MEDIUM VOLTAGE CABLES**

EDMS 28-400-1

Date: 21/11/2023

EDMS 28 - 400 -1

**SPECIFICATION FOR
HYDRAULIC CRIMPING TOOL FOR LOW AND MEDIUM
VOLTAGE CABLES**

TYPE 1 FROM 16 mm² UP TO 300 mm² (HAND TYPE)

TYPE 2 FROM 25 mm² UP TO 630 mm² (FOOT TYPE)

Issue: NOV -2023 / Rev-1

- This revision contains option items that must be selected by EDC before bidding.

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HYDRAULIC CRIMPING TOOL FOR LOW AND MEDIUM VOLTAGE CABLES
EDMS 28-400-1**Date: 21/11/2023****1. SCOPE:**

This specification covers the minimum technical requirements of the design, inspection, packing, marking, and performance of hydraulic cable crimping tools for low and medium voltage cables (AL/CU) as follows:

- Type1: from 16 mm² up to 300 mm² (hand type).
- Type2: from 25 mm² up to 630 mm² (foot type).

2. APPLICABLE CODES AND STANDARDS:

The hydraulic cable crimping tool covered in this specification should comply with the latest edition of IEC standards as given in Table (1).

Table (1)

Standard No.	Description
DIN 46329	AL Cable Lugs for Compression
DIN 46235	CU Cable Lugs for Compression
DIN 46267 part 1	CU Cable connectors
DIN 46267 part 2	AL Cable connectors
DIN 46243-Q	Accessories
DIN 48083	Hexagon crimping die

3. ENVIRONMENTAL CONDITIONS

The performance of the hydraulic cable crimping tool should be guaranteed under the environmental conditions given in Table (2).

Table (2)

Ambient temperature	-5°C to +45°C (50°C as option)
Maximum relative humidity	95 %
Altitude	≤ 1000 meters.

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4. DESIGN CRITERIA:

- The hydraulic cable crimping tool shall be of hexagonal crimping dies according to the latest editions of DIN standards for compression-type cable lugs and compression-type cable connectors and according to EDMS 15-300 & EDMS 15-301 technical specifications.
- The hydraulic cable crimping tool shall be suitable for crimping the compression-type cable lugs and compression-type cable connectors (AL/CU) with a cross-sectional area as follows:
 - Type1: from 16 mm² up to 300 mm² (hand type).
 - Type2: from 25 mm² up to 630 mm² (foot type).
- The hydraulic cable crimping tool shall be supplied with pre-rounding dies to convert the cable section from sector-shaped into a circular shape before crimping for all cable cross sections.
- The hydraulic cable crimping tool shall be supplied with hexagonal crimping dies at least as follows:
 - Type 1: (8-10-12-14-16-18-20-22-25-28-32-34)

Cross section (mm ²)	Copper Die code	Aluminum Die code
16	8	10
25	10	12
35	12	14
50	14	16
70	16	18
95	18	22
120	20	22
150	22	25
185	25	28
240	28	32
300	32	34

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- Type 2: (10-12-14-16-18-20-22-25-28-32-34-38-42-44- 46 or 52)

Cross section (m m ²)	Copper Die code	Aluminum Die code
25	10	12
35	12	14
50	14	16
70	16	18
95	18	22
120	20	22
150	22	25
185	25	28
240	28	32
300	32	34
400	38	38
500	42	44
630	44	46 or 52

- The dies should be of high-grade steel.
- The compression force of the hydraulic cable crimping tool shall be not less than :
 - Type 1: 10 tons (98 KN).
 - Type 2: 11 tons (107 KN).
- The weight of type 1 hydraulic cable crimping tool shall be not more than 7.7 kg.
- The hydraulic cable crimping tool head shall be axial and the axis of the center should rotate at least through 180 ° to use in narrow places.
- The hydraulic cable crimping tool should be equipped with an automatic pressure safety valve.
- The hydraulic cable crimping tool separates (self) once the complete pressing is completed.
- "C" shaped head with a large jaw or parallel design.
- The hand of the Hydraulic cable crimping tool should be electrically insulated.

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- The case of type 1 should be compact for hand use, lightweight, and easy to move provided with the crimper and the head with its dies.
- The supplier should provide a suitable box for a type 2 crimper with a foot-operated pump, an oil pressure gauge, and at least a five-meter flexible hose.
- The tenderer should supply nonconductive very high-pressure hose 3VE0 type with Compact size, low elongation, designed for permanent high-pressure couplings with hose guards, and lightweight.
- The hose data should be printed on it.
- The tenderer should supply a cutting head with cutting diameter not less than 12 cm (option as --EDC requirements).

5. SUBMITTALS

- A certificate of origin country and quality management shall be provided.
- Original catalogues in English shall be submitted to facilitate the evaluation of the offer.
- The tenderer should submit an instruction sheet in Arabic to explain the operation and maintenance steps.

6. MAINTENANCE:

The tenderer shall comply a free factory maintenance through the guarantee period.

7. PACKING AND MARKING:

- The hydraulic cable crimping tool shall be supplied complete with the crimping dies in a suitable handy anti-shock rigid case.
- The hydraulic cable crimping tool hexagonal dies shall be according to DIN standards and shall be marked according to them.

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1. The crimper shall be able to pass the tensile tests prescribed in IEC 61238-1 and the tensile force shall be as follows:

Class	Conductor Material	Nominal cross-sectional area A (mm ²)	Tensile force (N)
Class 1	Aluminum	≤ 500	$40 \times A$
		≥ 630	20000
	Copper	≤ 300	$60 \times A$
		≥ 400	20000

2. After the compression test: the dimension should be matched (cross sectional area $\times \sin 60^\circ$)

9. GUARANTEE

- The supplier shall guarantee the supplied hydraulic crimping tool against all defects arising out of faulty design, or of defective material for period of 18 month from the date of delivery.
- The supplier shall permanently supply spare parts for the maintenance of the crimping tool to --EDC as needed.

10. TRAINING

- The tenderer should offer a training on how to operate and maintain the crimper.

11. TECHNICAL DATA SCHEDULE

- The tenderer should fill in the attached technical data schedule.
- Any offer not accompanied by a clear and complete technical data schedule shall be rejected.

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TECHNICAL DATA SCHEDULE
HYDRAULIC CABLE CRIMPING TOOL

1- Maker's name	
2- Country of manufacture	
3- Crimper type	
4- Model No	
5- Application range (mm ²)	
6- Head shape	
7- Cutting head diameter (cm)	
8- Hose - Type - Length (m)	
9- The axis rotating degree °	
10- Die's material	
11- No of dies	
12- Standards dies	
13- Crimping force (ton)	
14- Arabic instruction sheet	
15- Weight (kg)	
16- Dimensions (L×W) (mm)	
17- Year of manufacturing	

I/we guarantee the correctness of the above data for hydraulic crimper we are offering to
Electricity Distribution Company

Signature: ----

Date: ----